

ARNAB DAS

B.Tech, Materials Science & Engineering, Indian Institute of Technology Delhi

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EDUCATION

Indian Institute of Technology Delhi

2024 – 2028

B.Tech in Materials Science & Engineering · CGPA 8.78 / 10 · Departmental Rank 2

Merit Scholarship, top 7% of 1,200 students (2025) · Fall Exchange Programme, IMT Mines Alès, France (2026) · JEE Main 99.84 percentile

SELECTED PROJECTS

NanoVector: Vector Database & Retrieval Engine

2026

C++17 · Python · pybind11 · FastAPI · AWS EC2 · GitHub Actions · github.com/arnabdas1034/NanoVector

- Implemented a vector database in C++ from first principles with **flat**, **KD-tree** and **HNSW** index backends, exposed to Python through pybind11 rather than depending on a managed vector store.
- Built a **retrieval evaluation harness** measuring Recall@k, MRR and p50/p95 latency, and used it to drive index decisions; deployed on AWS EC2 behind FastAPI, with C++ and pytest suites in CI.

Ember: Desktop Client for API-Key Users

2026

TypeScript · Anthropic API · github.com/arnabdas1034/ember

- Students on YC and startup programmes receive large model-API credit grants, but the only way to spend them is to write code against an API. There is no consumer-grade chat product that runs on your own key. **I had the problem myself, then asked around and found everyone assumed such a thing already existed.**
- Building that product: a desktop chat client that turns an API key into something that behaves like a paid subscription, covering streaming, persistent history and message management, with no credentials held by the app.

ColdEmail: Outreach Automation Platform

2026

TypeScript · LLM APIs · GitHub Actions · github.com/arnabdas1034/ColdEmail

- Built and deployed an end-to-end outreach system that sources contacts, verifies deliverability before send, and **generates a personalised opener per lead** through an LLM. Roughly 5,000 lines across 44 commits, with CI from the first week.

GrainSim: Materials Simulation Service

2026

Python · pytest · Docker · github.com/arnabdas1034/GrainSim

- Wrote a materials physics simulation spanning **16 material systems**, my degree subject written as code I could run, with a real pytest suite and a containerised backend.

Currently building: an autonomous code-review agent on AWS Lambda with GitHub PR integration; a URL shortener with analytics; a concurrent job scheduler in C++17; and a campus visitor-entry and QR gate-ticketing system commissioned through IIT Delhi's Security Head, prototype due August 2026.

OPEN SOURCE

Valhalla Routing Engine: merged contribution, PR #6159

Jul 2026

C++ · CMake · cross-platform CI

- Landed a **maintainer-reviewed and merged** change in Valhalla, an open-source routing engine with 6,000+ GitHub stars used in production routing stacks.
- Added a configure-time dependency check so a missing *spatialite-tools* fails immediately with a clear diagnostic, rather than breaking opaquely later during timezone database generation. This **closed an issue open for four years** (#2202).

TECHNICAL SKILLS

Languages: C++17, Python, TypeScript / JavaScript, SQL

Backend & AI: FastAPI, NestJS, REST APIs, RAG and vector retrieval, LLM APIs and tool calling, pybind11, PostgreSQL, pgvector, Redis

Infrastructure: AWS (EC2, S3, Lambda), Docker, GitHub Actions, CMake, Git, Linux

LEADERSHIP

Secretary, Student Affairs Council · Karakoram Hostel, IIT Delhi

Jan 2026 – Jan 2027

- Represent the hostel to the Warden, BHM and institute administration on student welfare, events and budgets. Automated council account provisioning with a scripted DOM filler and a Python generation and de-duplication pass, **creating 62 administrative accounts** that would otherwise have been entered by hand.

Previously Camp Manager, BloodConnect (NSS IIT Delhi), running blood donation camps end to end, 2025–26.